NE-0b residual verification notes (BG-NASEMU) — 2026-11-13 ~00:32-00:45 CET

Read-only; no sudo, no mounts, no pulls, no VM run. Raw artifacts alongside: fs.pdf (File Station API Guide, 5 PDF "pages" but 6141 text lines via pdftotext), fs.txt, web_auth.html/.txt, nfs.html.

1. Synology API contract (source: Synology_File_Station_API_Guide.pdf from cndl.synology.cn, fetched 2026-11-13, 502145 B)

Doc is "Synology File Station Official API", marked proprietary (no reproduction/design-basis without permission): contract notes below are paraphrased minimal facts for interop; legal use as shim basis = UNCONFIRMED/operator question. NOTE: URL https://help.synology.com/developer-guide/integrate_dsm/web_authentication.html is actually "Application Authentication" (package CGI authenticate.cgi), NOT the Login Web API. It only confirms login URL form `/webapi/auth.cgi?api=SYNO.API.Auth&version=3&method=login&account=..&passwd=..&format=cookie`, ports 5001 https / 5000 http, logout `version=1&method=logout`. The separate "DSM Login Web API" PDF was NOT found/opened (UNCONFIRMED). Request form: GET /webapi/?api=&version=&method=[&params][&_sid=] (or cookie `id`); params escaped (comma->\... per doc); passwd not escaped; POST also accepted for upload. Response: JSON {"success":true,"data":{...}} or {"success":false,"error":{"code":N[,"errors":[...]]}}.

- SYNO.API.Info v1, since DSM 4.0, fixed at /webapi/query.cgi, method=query&query=all|comma-list -> data{:{path,minVersion,maxVersion}}. Doc examples: SYNO.API.Auth path auth.cgi min1 max3; SYNO.FileStation.List path FileStation/file_share.cgi (old v1 example); File Station v2 APIs (DSM 6.0+) use /webapi/entry.cgi. No API-specific error codes.
- SYNO.API.Auth: v3 since DSM 4.2, v2 since 4.1. login params: account, passwd, session (e.g. FileStation), format=cookie|sid (v2+), otp_code (v3, reserved; "WebAPI doesn't support it yet"). Response data.sid. sid expires 7 days default. logout params: session; empty success. Errors: 400 no such account/incorrect password; 401 account disabled; 402 permission denied; 403 2-step code required; 404 2-step code failed.
- Common error codes: 100 unknown; 101 no API/method/version param; 102 API does not exist; 103 method does not exist; 104 version unsupported for functionality; 105 session lacks permission; 106 session timeout; 107 session interrupted by duplicate login; 119 SID not found. File-op common: 400 invalid param; 401 unknown; 402 system busy; 403 invalid user; 404 invalid group; 405 invalid user and group.
- SYNO.FileStation.Info v2 (DSM 6.0+) method get (no params), entry.cgi -> data{is_manager,support_virtual_protocol e.g. "cifs,nfs,iso",support_sharing,hostname}.
- SYNO.FileStation.List v2: list_share (offset,limit,sort_by,sort_direction,onlywritable,additional[real_path,owner,time,perm,mount_point_type,sync_share,volume_status]) -> {total,offset,shares[{path,name,additional}]}; list (folder_path req; offset,limit,sort_by,sort_direction,pattern,filetype file|dir|all,goto_path,additional[real_path,size,owner,time,perm,type]) -> {total,offset,files[{path,name,isdir,additional{size,time{atime,mtime,ctime,crtime seconds},owner{user,group,uid,gid},perm{posix,...},type=extension}}]}; getinfo (path JSON-array string) -> files[]. additional passed as JSON array string e.g. ["real_path","size"]. (isdir field inferred from doc object; exact field table partially read: UNCONFIRMED line-level.)
- SYNO.FileStation.Download v2 method download: path=["/share/f.txt"] (JSON array, URL-encoded), mode=open|download; one file -> raw content, many -> ZIP; no API error codes; with mode=open errors give HTTP 404.
- SYNO.FileStation.Upload v2 method upload: POST /webapi/entry.cgi multipart/form-data (RFC 1867), parts api,version,method,path (dest folder starting with share),create_parents,overwrite(true/false; v3 overwrite|skip),mtime/crtime/atime (ms), file part LAST. Success: empty success. Errors: 1800 no/mismatched Content-Length; 1801 client wait timeout (3600s); 1802 no filename in last part; 1803 connection cancelled; 1804 oversized file to FAT; 1805 cannot overwrite/skip existing file when no overwrite param. Shim implication: minimal set = Info.query, Auth.login/logout (+ error 400/402/105/106/119), FileStation Info.get, List.list_share/list/getinfo, Download.download, Upload.upload. 2FA(403/404) not testable via official WebAPI per doc.

2. Host facts (2026-11-13)

- /dev/sda1: 7.3T, FSTYPE ntfs, LABEL LOCAL7, UUID 6EACEE6FACEE316F, device root:disk 0660 (user not in disk group). `findmnt -S /dev/sda1` empty => NOT mounted. Not writable by us without sudo/udisks mount; NTFS is a poor VM image store (no sparse/perm semantics, ntfs3/ntfs-3g needed). Safe? Not usable rootless today; needs operator mount decision. Contents/ownership UNKNOWN (not mounted, not inspected).
- `/` and $HOME: /dev/nvme1n1p2 ext4 1.8T used 1.7T avail 55G (97%). Only 55G free; vdsm (>=16G) + dockur/windows (>=32G) + macos (64G default) cannot all fit.
- RAM: MemTotal 31717516 kB (~30.2 GiB), MemAvailable 20163880 kB (~19.2 GiB) at 00:32.

3. Rootless prerequisites

- podman rootless=true, cgroup v2 (systemd manager), netavark 1.16.1, aardvark-dns 1.16.0, crun 1.21, rootlessNetworkCmd=pasta (pasta 0~git20260120), slirp4netns also present, buildah 1.42.1; Ubuntu 26.04.1.
- /dev/kvm crw-rw----+ root:kvm, ACL user:milosvasic:rw- ; in-container `--device /dev/kvm` with alpine:3.20: listed as nobody:nobody, `test -r -a -w` => kvm-rw-ok. KVM usable rootless.
- /dev/net/tun crw-rw-rw- : `podman run --rm --cap-add NET_ADMIN --device /dev/net/tun docker.io/library/alpine:3.20` rc=0; in-container open of tun fd OK ("tun-open-ok"). (First attempt with name localhost/alpine:3.20 failed only because wrong image name: it tried to pull, failed; local image is docker.io/library/alpine:3.20.)
- /dev/vhost-net crw-rw---- root:kvm, NO ACL for user, user not in kvm group => NOT accessible: `--device /dev/vhost-net` starts (rc=0) but open inside fails "Permission denied". => dockur/vdsm DHCP/macvlan modes needing vhost-net unavailable; default NAT/user-mode networking is the route (needs no vhost). Fix would be `setfacl -m u:milosvasic:rw /dev/vhost-net` or kvm group = sudo, operator action.
- net.ipv4.ip_unprivileged_port_start=1024 (ports <1024 need remap); user.max_user_namespaces=123694; ulimit -u 123694; subuid milosvasic:100000:65536.
- Not tested: actually running vdsm/dockur (no image, no pull, per scope). vdsm rootless boot result still UNCONFIRMED; verdict so far: prerequisites kvm+tun OK, vhost-net blocked, disk-space tight.

4. Licences / releases (GitHub API 2026-11-13 unless noted)

- rclone MIT, v1.76.0 2026-11-06, pushed 2026-11-12. `rclone serve nfs` (rclone.org/commands/rclone_serve_nfs): NFSv3 only; NO authentication (any client); default random port on loopback, use --addr 0.0.0.0:PORT to expose; writes need --vfs-cache-mode (full recommended) else read-only; mount with `-o port=$PORT,mountport=$PORT,tcp`; --nfs-cache-type memory (default; restart => stale handles) | disk | symlink (Linux only, needs root or CAP_DAC_READ_SEARCH => unusable rootless); --nfs-cache-handle-limit default 1000000; subdir mount supported.
- dufs Apache-2.0, v0.47.0 2026-10-17, pushed 2026-11-10.
- unfs3 licence NOASSERTION per GitHub (BSD-like per plan, UNCONFIRMED; LICENSE file not read), v0.11.0 2025-03-28, pushed 2025-07-03, NFSv3 userspace.
- ProFTPD GPL-2.0, v1.3.9e 2026-10-26, pushed 2026-11-12.
- nginx BSD-2-Clause, release-1.31.8 2026-10-21 (note: nginx core has built-in ngx_http_dav_module with PUT/DELETE/MKCOL/COPY/MOVE only, no PROPFIND — full WebDAV needs 3rd-party ext module; from memory, UNCONFIRMED here). Apache mod_dav: not checked (UNCONFIRMED).
- vsftpd: primary page security.appspot.com/vsftpd.html: latest news 3.0.5 (Aug 2021; 3.0.4/3.0.5 after 6 years gap). Licence text not shown on that page (GPLv2 per common knowledge, UNCONFIRMED from primary). Low-activity.

5. Remaining unresearched platforms

- Unraid: proprietary paid OS (unraid.net/pricing: perpetual licence, 30-day free trial, "Docker containers" and "Virtual Machines" are features OF Unraid, i.e. it is a host OS, not a container). Cannot be run as a container; VM-only, licence/trial needed. Exclude.
- TrueNAS SCALE/CE: full OS installed from ISO on hardware or VM (truenas.com install docs); apps catalog truenas/apps is LGPL-3.0 (compose catalog, not the NAS itself). VM-only. Optional VM-tier.
- OpenMediaVault: official docs state verbatim "openmediavault does not support installation in LXC or any other container based solution." => VM-only (or Debian host install).

- CasaOS: Apache-2.0, last release v0.4.15 2024-12-19; home-server app UI, not a NAS protocol emulator. Exclude.